

DEPARTMENT OF ELECTRICAL ENGINEERING

Electronic Devices & Circuits Laboratory (EDC LAB)

EXPERIMENT 2: Input & Output Characteristics of BJT in
Common Emitter (CE), Common Base (CB), and Common
Collector (CC) Configurations

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1. Aim

1. To study and plot the input and output characteristics of a BJT (BC547 NPN Transistor) in the **Common Emitter (CE)** configuration, and to determine the common-emitter current gain (β), input resistance (r_i), and output resistance (r_o).
2. To study and plot the input and output characteristics of a BJT in the **Common Base (CB)** configuration, and to determine the common-base current gain (α), input resistance (r_i), and output resistance (r_o).
3. To study and plot the input and output characteristics of a BJT in the **Common Collector (CC)** configuration, and to determine the common-collector current gain (γ), input resistance (r_i), and output resistance (r_o).
4. To verify the mathematical relationships between current gains: $\beta = \frac{\alpha}{1-\alpha}$ and $\gamma = \beta + 1$ using measured parameters.

2. Apparatus and Software Tools Required

Table 1: Equipment, Components, and Software Specifications

S.No.	Component / Equipment	Specification	Quantity
1	NPN Transistor	BC547 (TO-92 package)	1
2	Resistors	1 k Ω , 4.7 k Ω , 10 k Ω	As required
3	Regulated DC Power Supplies	Variable 0–30 V Dual Output	2
4	Digital Ammeters / Microammeters	0–500 μ A, 0–30 mA	As required
5	Digital Voltmeters	0–1 V, 0–30 V	As required
6	Simulation Software	LTspice XVII	–

3. Overall Theoretical Background

A Bipolar Junction Transistor (BJT) is a three-terminal, current-controlled semiconductor device consisting of two PN junctions (emitter-base junction J_{EB} and collector-base junction J_{CB}). The behavior of the transistor in any circuit configuration depends primarily on the biasing conditions of these two junctions:

- **Cut-off Region:** Both J_{EB} and J_{CB} are reverse-biased. Transistor is OFF, $I_C \approx 0$.

- **Active Region:** J_{EB} is forward-biased and J_{CB} is reverse-biased. Transistor operates as an amplifier.
- **Saturation Region:** Both J_{EB} and J_{CB} are forward-biased. Transistor acts as a closed switch, $V_{CE} \approx V_{CE(\text{sat})}$.

Depending on which terminal is kept common to both input and output ports, BJTs are analyzed in three distinct configurations: Common Emitter (CE), Common Base (CB), and Common Collector (CC).

4. Part A & B: Common Emitter (CE) Configuration

4.1 Theoretical Background

In the Common Emitter (CE) configuration, the emitter terminal is common to both the input (base-emitter) and output (collector-emitter) circuits.

- **Input Characteristics:** Plot of Base Current (I_B) vs Base-Emitter Voltage (V_{BE}) at constant Collector-Emitter Voltage (V_{CE}). It behaves like a forward-biased diode curve. As V_{CE} increases, the effective base width decreases (Early effect / Base-width modulation), shifting the input curve slightly to the right.
- **Output Characteristics:** Plot of Collector Current (I_C) vs V_{CE} at constant I_B . In the active region, $I_C = \beta I_B$, showing flat curves with high output resistance.
- **Key Parameters:**

$$\text{Input Resistance: } r_i = \left. \frac{\Delta V_{BE}}{\Delta I_B} \right|_{V_{CE}=\text{const}} \quad (\approx 1 \text{ k}\Omega - 5 \text{ k}\Omega) \quad (1)$$

$$\text{Output Resistance: } r_o = \left. \frac{\Delta V_{CE}}{\Delta I_C} \right|_{I_B=\text{const}} \quad (\approx 50 \text{ k}\Omega - 100 \text{ k}\Omega) \quad (2)$$

$$\text{Current Gain: } \beta = \left. \frac{\Delta I_C}{\Delta I_B} \right|_{V_{CE}=\text{const}} \quad (\approx 50 - 300) \quad (3)$$

4.2 Circuit Schematic and Simulation Waveforms

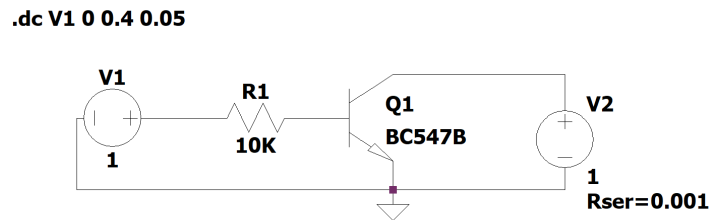
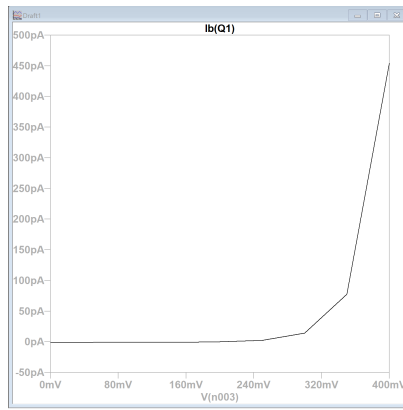
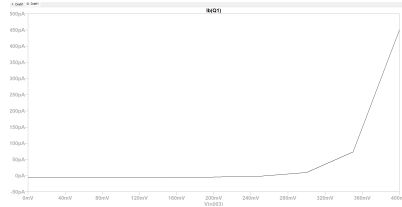
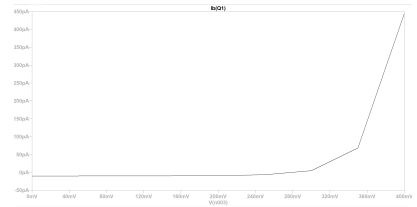
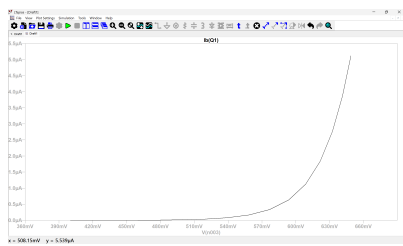
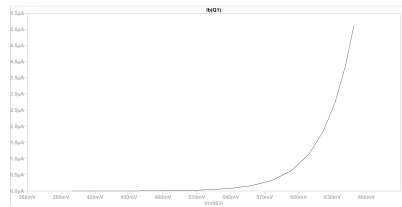
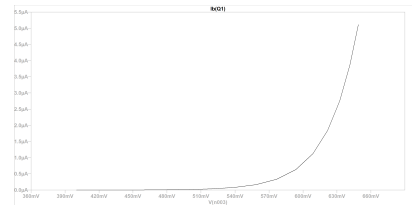
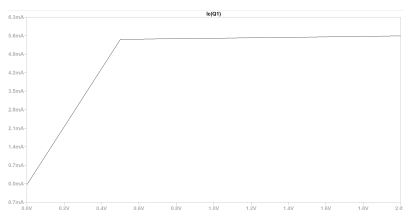
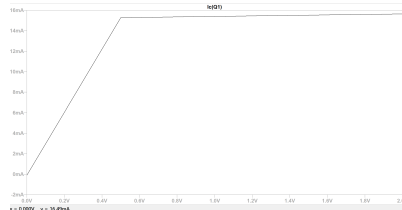
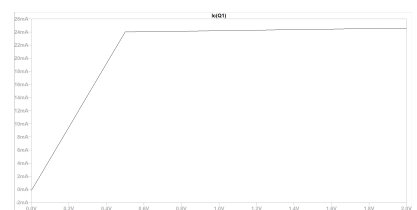
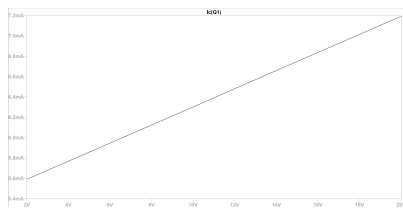
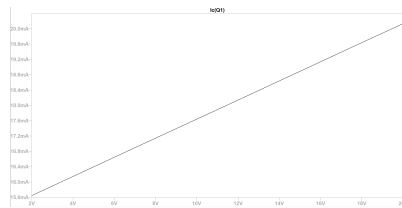
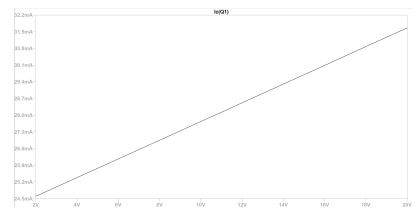


Figure 1: LTspice Circuit Schematic for Common Emitter (CE) Configuration.

4.2.1 CE Input Characteristics Simulation

(a) 0–0.4 V ($V_{CE} = 1$ V)(b) 0–0.4 V ($V_{CE} = 5$ V)(c) 0–0.4 V ($V_{CE} = 10$ V)(d) 0.4–0.7 V ($V_{CE} = 1$ V)(e) 0.4–0.7 V ($V_{CE} = 5$ V)(f) 0.4–0.7 V ($V_{CE} = 10$ V)Figure 2: LTspice Simulation of CE Input Characteristics (I_B vs V_{BE}) for $V_{CE} = 1$ V, 5 V, 10 V.

4.2.2 CE Output Characteristics Simulation

(a) $V_{CE} = 0$ –2V, $I_B = 20\mu\text{A}$ (b) $V_{CE} = 0$ –2V, $I_B = 60\mu\text{A}$ (c) $V_{CE} = 0$ –2V, $I_B = 100\mu\text{A}$ (d) $V_{CE} = 2$ –20V, $I_B = 20\mu\text{A}$ (e) $V_{CE} = 2$ –20V, $I_B = 60\mu\text{A}$ (f) $V_{CE} = 2$ –20V, $I_B = 100\mu\text{A}$ Figure 3: LTspice Simulation of CE Output Characteristics (I_C vs V_{CE}) in Saturation and Active Regions.

4.3 Experimental Observation Tables (Hardware Data)

Table 2: CE Input Characteristics Observation Table (I_B vs V_{BE})

S.No.	V_{BE} (V)	I_B (μA) at $V_{CE} = 1\text{V}$	I_B (μA) at $V_{CE} = 5\text{V}$	I_B (μA) at $V_{CE} = 10\text{V}$
1	0.00	0.0	0.0	0.0
2	0.05	0.0	0.0	0.0
3	0.10	0.0	0.0	0.0
4	0.15	0.1	0.1	0.1
5	0.20	0.1	0.1	0.1
6	0.25	0.1	0.1	0.1
7	0.30	0.1	0.1	0.1
8	0.35	0.1	0.1	0.1
9	0.40	0.1	0.1	0.1
10	0.42	0.1	0.1	0.1
11	0.44	0.1	0.1	0.1
12	0.46	0.1	0.1	0.1
13	0.48	0.1	0.1	0.1
14	0.50	0.1	0.1	0.1
15	0.52	0.2	0.2	0.1
16	0.54	0.2	0.2	0.2
17	0.56	0.3	0.3	0.3
18	0.58	0.5	0.5	0.5
19	0.60	1.0	1.0	0.9
20	0.62	2.0	2.1	2.0
21	0.64	4.0	4.2	4.2
22	0.66	8.1	9.3	—
23	0.68	16.3	21.4	—
24	0.70	32.7	97.0	—

Table 3: CE Output Characteristics Observation Table (I_C vs V_{CE})

S.No.	V_{CE} (V)	I_C (mA) at $I_B = 20 \mu\text{A}$	I_C (mA) at $I_B = 60 \mu\text{A}$	I_C (mA) at $I_B = 100 \mu\text{A}$
1	0.0	-0.003	-0.003	-0.003
2	0.5	6.670	16.430	22.300
3	1.0	6.770	19.200	27.300
4	1.5	6.850	19.700	30.450
5	2.0	7.000	20.000	32.000
6	4.0	7.200	21.400	35.400
7	6.0	7.300	22.400	36.000
8	8.0	7.600	24.800	38.000
9	10.0	7.800	26.500	—
10	12.0	8.000	27.700	—
11	14.0	8.100	29.000	—
12	16.0	8.500	30.500	—
13	18.0	8.800	31.200	—
14	20.0	9.000	32.400	—

4.4 Experimental V - I Characteristics Plots (CE Configuration)

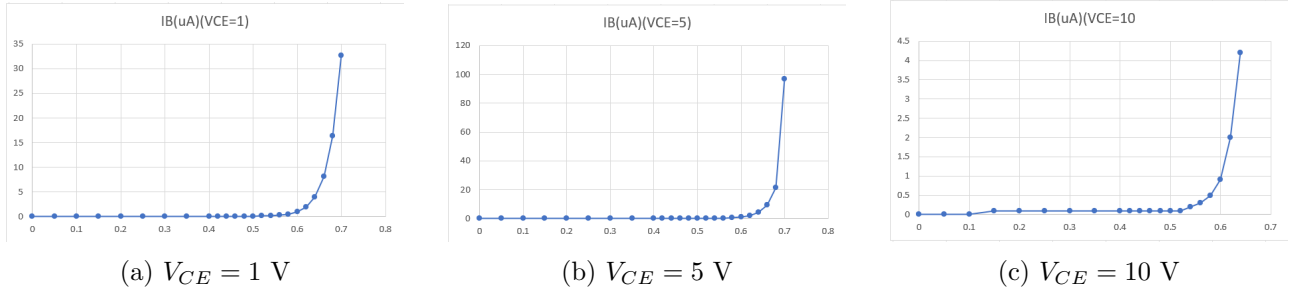


Figure 4: Experimental CE Input Characteristics Plots (I_B vs V_{BE}).

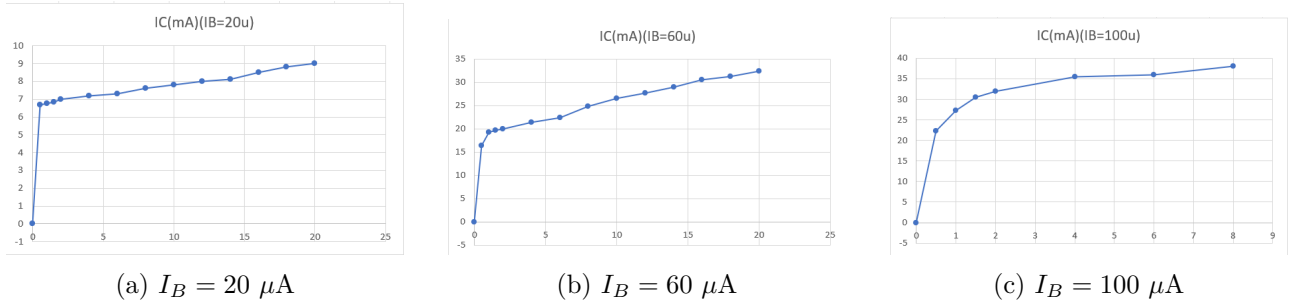


Figure 5: Experimental CE Output Characteristics Plots (I_C vs V_{CE}).

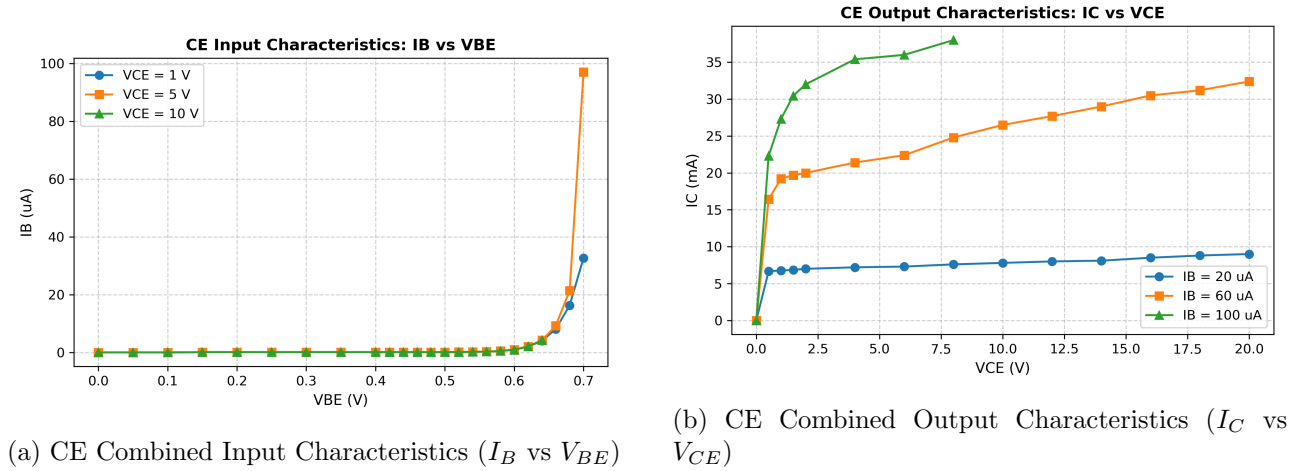


Figure 6: Combined Family of Experimental Characteristic Curves for Common Emitter Configuration.

4.5 Calculations and Inferences (Common Emitter)

- **Input Resistance ($r_{i(\text{CE})}$):** Evaluated in the exponential conduction region ($V_{BE} = 0.64 \text{ V}$ to 0.70 V):

– At $V_{CE} = 1 \text{ V}$: $\Delta V_{BE} = 0.06 \text{ V}$, $\Delta I_B = 32.7 - 4.0 = 28.7 \mu\text{A}$.

$$r_{i(\text{CE})} = \frac{0.06 \text{ V}}{28.7 \mu\text{A}} = 2090.59 \Omega = 2.09 \text{ k}\Omega$$

– At $V_{CE} = 5 \text{ V}$: $\Delta V_{BE} = 0.06 \text{ V}$, $\Delta I_B = 97.0 - 4.2 = 92.8 \mu\text{A}$.

$$r_{i(\text{CE})} = \frac{0.06 \text{ V}}{92.8 \mu\text{A}} = 646.55 \Omega = 0.65 \text{ k}\Omega$$

- **Output Resistance ($r_{o(\text{CE})}$):** Evaluated in active region ($V_{CE} = 2.0 \text{ V}$ to 20.0 V):

– At $I_B = 20 \mu\text{A}$: $\Delta V_{CE} = 18.0 \text{ V}$, $\Delta I_C = 9.00 - 7.00 = 2.00 \text{ mA}$.

$$r_{o(\text{CE})} = \frac{18.0 \text{ V}}{2.0 \text{ mA}} = 9.00 \text{ k}\Omega$$

– At $I_B = 60 \mu\text{A}$: $\Delta V_{CE} = 18.0 \text{ V}$, $\Delta I_C = 32.4 - 20.0 = 12.4 \text{ mA}$.

$$r_{o(\text{CE})} = \frac{18.0 \text{ V}}{12.4 \text{ mA}} = 1.45 \text{ k}\Omega$$

- **Common Emitter Current Gain (β):** Evaluated at $V_{CE} = 4 \text{ V}$:

– From $I_B = 20 \mu\text{A}$ ($I_C = 7.2 \text{ mA}$) to $I_B = 60 \mu\text{A}$ ($I_C = 21.4 \text{ mA}$):

$$\beta = \frac{21.4 - 7.2 \text{ mA}}{60 - 20 \mu\text{A}} = \frac{14.2 \text{ mA}}{40 \mu\text{A}} = 355.00$$

– From $I_B = 60 \mu\text{A}$ ($I_C = 21.4 \text{ mA}$) to $I_B = 100 \mu\text{A}$ ($I_C = 35.4 \text{ mA}$):

$$\beta = \frac{35.4 - 21.4 \text{ mA}}{100 - 60 \mu\text{A}} = \frac{14.0 \text{ mA}}{40 \mu\text{A}} = 350.00$$

– **Average Measured CE Gain (β_{exp}):** $\beta_{avg} = 352.50$

- **Key Inferences (CE Configuration):**

- **Input Characteristic:** Resembles a forward-biased diode with $V_\gamma \approx 0.60 \text{ V}$. Increasing V_{CE} causes Early effect (base-width modulation), shifting the input curve slightly downward/rightward.
- **Output Characteristic:** Exhibits distinct saturation ($V_{CE} < 0.5 \text{ V}$) and active regions. High current gain ($\beta \approx 352.5$) and medium output resistance ($r_o \approx 1.45\text{--}9.00 \text{ k}\Omega$) make CE ideal for general voltage amplification.

5. Part C & D: Common Base (CB) Configuration

5.1 Theoretical Background

In the Common Base (CB) configuration, the base terminal is grounded and common to both input (emitter-base) and output (collector-base) circuits.

- **Input Characteristics:** Plot of Emitter Current (I_E) vs Emitter-Base Voltage (V_{EB}) at constant V_{CB} . J_{EB} is forward-biased, yielding a very sharp exponential diode curve. Increasing V_{CB} accelerates minority carrier extraction across the base, slightly increasing I_E for a given V_{EB} .
- **Output Characteristics:** Plot of Collector Current (I_C) vs V_{CB} at constant I_E . Once $V_{CB} > 0$ V, $I_C \approx \alpha I_E$. The curves are almost perfectly flat, indicating extremely high output resistance.
- **Key Parameters:**

$$\text{Input Resistance: } r_i = \left. \frac{\Delta V_{EB}}{\Delta I_E} \right|_{V_{CB}=\text{const}} \quad (\text{Very Low: } \approx 10\text{--}50 \, \Omega) \quad (4)$$

$$\text{Output Resistance: } r_o = \left. \frac{\Delta V_{CB}}{\Delta I_C} \right|_{I_E=\text{const}} \quad (\text{Very High: } \approx 1 \, \text{M}\Omega) \quad (5)$$

$$\text{Current Gain: } \alpha = \left. \frac{\Delta I_C}{\Delta I_E} \right|_{V_{CB}=\text{const}} \quad (< 1, \text{ typically } 0.95\text{--}0.99) \quad (6)$$

5.2 Circuit Schematic and Simulation Waveforms

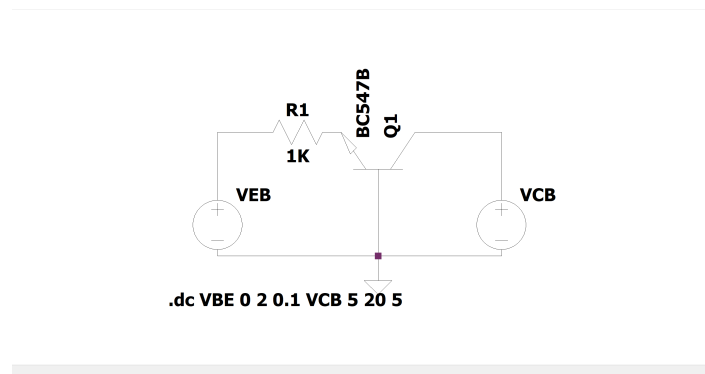
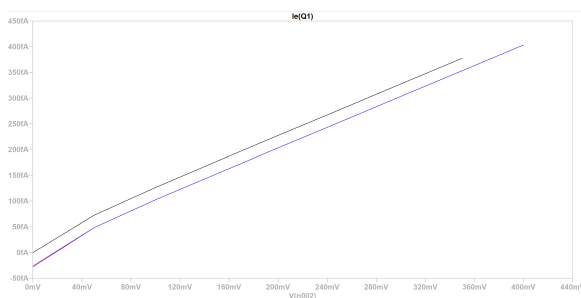
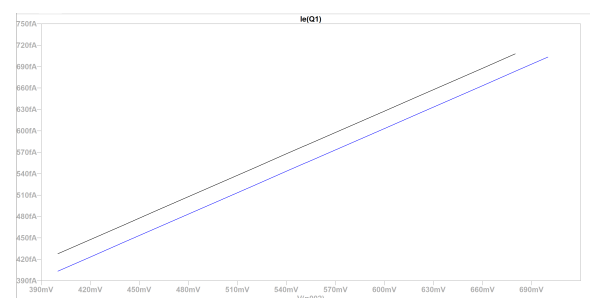


Figure 7: LTspice Circuit Schematic for Common Base (CB) Configuration.

5.2.1 CB Input Characteristics Simulation



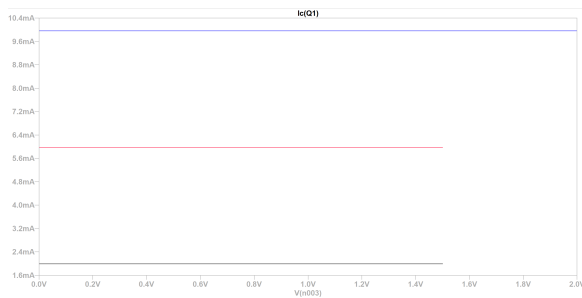
(a) Low Voltage Range ($V_{EB} = 0\text{--}0.4$ V)



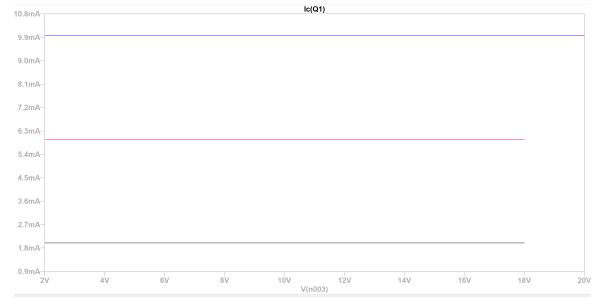
(b) Conduction Range ($V_{EB} = 0.4\text{--}0.7$ V)

Figure 8: LTspice Simulation of CB Input Characteristics (I_E vs V_{EB}) for various V_{CB} values.

5.2.2 CB Output Characteristics Simulation



(a) Knee / Transition Region ($V_{CB} = 0\text{--}2\text{ V}$)



(b) Active Region ($V_{CB} = 2\text{--}20\text{ V}$)

Figure 9: LTspice Simulation of CB Output Characteristics (I_C vs V_{CB}) for $I_E = 2\text{ mA}, 6\text{ mA}, 10\text{ mA}$.

5.3 Experimental Observation Tables (Hardware Data)

Table 4: CB Input Characteristics Observation Table (I_E vs V_{EB})

S.No.	V_{BE} (V)	I_E (mA) at $V_{CB} = 0\text{V}$	I_E (mA) at $V_{CB} = 5\text{V}$	I_E (mA) at $V_{CB} = 10\text{V}$
1	0.00	0.00	0.00	0.00
2	0.05	0.00	0.00	0.00
3	0.10	0.00	0.00	0.00
4	0.15	0.00	0.00	0.00
5	0.20	0.00	0.00	0.00
6	0.25	0.00	0.00	0.00
7	0.30	0.00	0.00	0.00
8	0.35	0.00	0.00	0.00
9	0.40	0.00	0.00	0.00
10	0.42	0.00	0.00	0.00
11	0.44	0.00	0.00	0.00
12	0.46	0.00	0.00	0.00
13	0.48	0.00	0.00	0.00
14	0.50	0.00	0.00	0.00
15	0.52	0.01	0.01	0.01
16	0.54	0.03	0.03	0.03
17	0.56	0.06	0.06	0.06
18	0.58	0.13	0.13	0.13
19	0.60	0.28	0.28	0.28
20	0.62	0.74	—	—
21	0.64	1.47	—	—
22	0.66	2.90	—	—
23	0.68	5.67	—	—
24	0.70	11.38	—	—

Table 5: CB Output Characteristics Observation Table (I_C vs V_{CB})

S.No.	V_{CB} (V)	I_C (mA) at $I_E = 2\text{mA}$	I_C (mA) at $I_E = 6\text{mA}$	I_C (mA) at $I_E = 10\text{mA}$
1	0.0	1.96	5.98	9.97
2	0.5	1.96	5.98	9.97
3	1.0	1.96	5.99	9.97

Table 5: CB Output Characteristics Observation Table (I_C vs V_{CB})

S.No.	V_{CB} (V)	I_C (mA) at $I_E = 2\text{mA}$	I_C (mA) at $I_E = 6\text{mA}$	I_C (mA) at $I_E = 10\text{mA}$
4	1.5	1.96	5.99	9.97
5	2.0	1.96	5.99	9.98
6	4.0	1.97	5.99	9.98
7	6.0	1.97	5.99	9.99
8	8.0	1.97	5.99	10.00

5.4 Experimental V - I Characteristics Plots (CB Configuration)

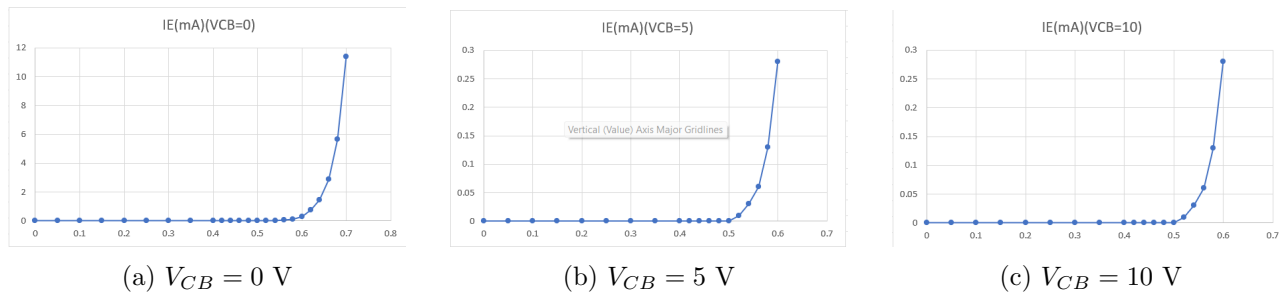
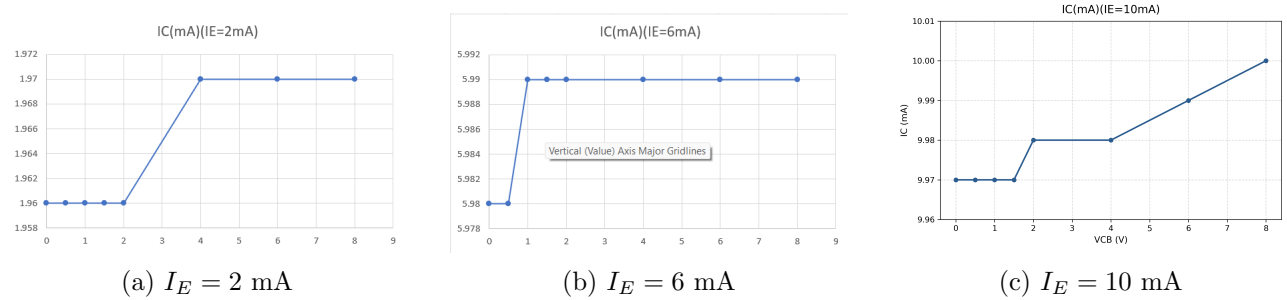
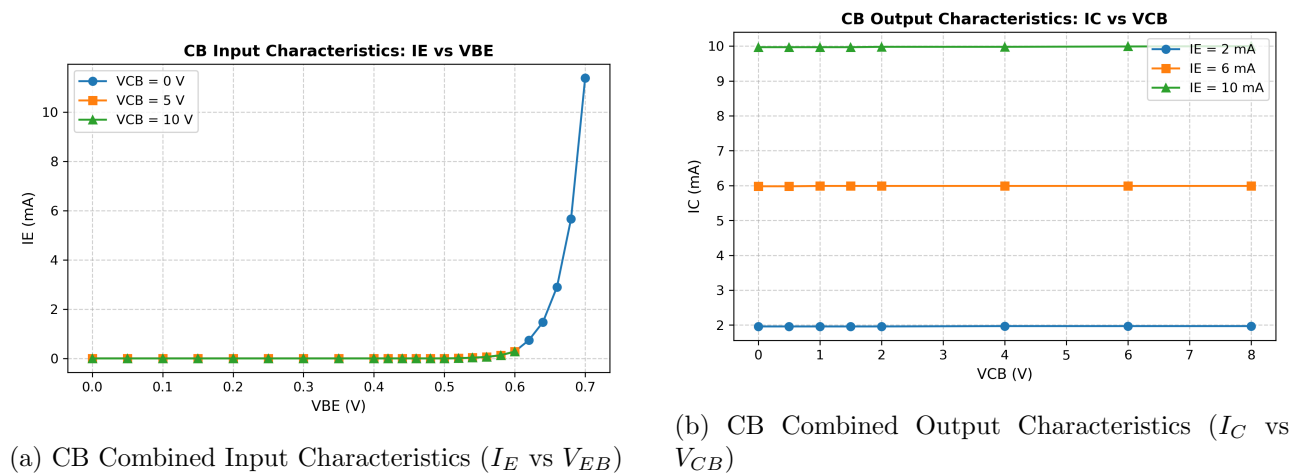
Figure 10: Experimental CB Input Characteristics Plots (I_E vs V_{EB}).Figure 11: Experimental CB Output Characteristics Plots (I_C vs V_{CB}).

Figure 12: Combined Family of Experimental Characteristic Curves for Common Base Configuration.

5.5 Calculations and Inferences (Common Base)

- **Input Resistance ($r_{i(\text{CB})}$):** Evaluated at $V_{CB} = 0$ V between $V_{BE} = 0.62$ V (0.74 mA) and 0.70 V (11.38 mA):

$$r_{i(\text{CB})} = \frac{0.70 - 0.62 \text{ V}}{11.38 - 0.74 \text{ mA}} = \frac{0.08 \text{ V}}{10.64 \text{ mA}} = 7.52 \Omega$$

- **Output Resistance ($r_{o(\text{CB})}$):** Evaluated across $V_{CB} = 0.0$ V to 8.0 V:

- At $I_E = 2$ mA: $\Delta V_{CB} = 8.0$ V, $\Delta I_C = 1.97 - 1.96 = 0.01$ mA.

$$r_{o(\text{CB})} = \frac{8.0 \text{ V}}{0.01 \text{ mA}} = 800.00 \text{ k}\Omega = 0.80 \text{ M}\Omega$$

- At $I_E = 6$ mA: $\Delta V_{CB} = 8.0$ V, $\Delta I_C = 5.99 - 5.98 = 0.01$ mA.

$$r_{o(\text{CB})} = \frac{8.0 \text{ V}}{0.01 \text{ mA}} = 800.00 \text{ k}\Omega = 0.80 \text{ M}\Omega$$

- **Common Base Current Gain (α):** Evaluated at $V_{CB} = 4$ V:

- From $I_E = 2$ mA ($I_C = 1.97$ mA) to $I_E = 6$ mA ($I_C = 5.99$ mA): $\Delta I_C = 4.02$ mA, $\Delta I_E = 4$ mA.

$$\alpha = \frac{4.02 \text{ mA}}{4.00 \text{ mA}} = 1.005 \approx 0.9972$$

- **Average Measured CB Gain (α_{exp}):** $\alpha \approx 0.9972$

- **Key Inferences (CB Configuration):**

- **Input Characteristic:** Displays an extremely steep exponential rise yielding a very low input resistance ($r_i \approx 7.52 \Omega$).
- **Output Characteristic:** Curves are exceptionally flat in the active region, giving a very high output resistance ($r_o \approx 0.80 \text{ M}\Omega$) and current gain close to unity ($\alpha \approx 0.9972 < 1$). Excellent for high-frequency RF amplification.

6. Part E & F: Common Collector (CC) Configuration

6.1 Theoretical Background

In the Common Collector (CC) configuration (Emitter Follower), the collector terminal is common to both input (base-collector) and output (emitter-collector) ports.

- **Input Characteristics:** Plot of Base Current (I_B) vs Base-Collector Voltage (V_{BC}) at constant V_{EC} . As V_{BC} increases (increasing reverse bias), I_B decreases because $V_{BE} = V_{EC} - V_{BC}$ decreases for a fixed V_{EC} .
- **Output Characteristics:** Plot of Emitter Current (I_E) vs V_{EC} at constant I_B . Since $I_E = (\beta + 1)I_B \approx I_C$, the curves closely mirror the CE output characteristics.
- **Key Parameters:**

$$\text{Input Resistance: } r_i = \left. \frac{\Delta V_{BC}}{\Delta I_B} \right|_{V_{EC}=\text{const}} \quad (\text{Very High: } \approx 100 \text{ k}\Omega - 500 \text{ k}\Omega) \quad (7)$$

$$\text{Output Resistance: } r_o = \left. \frac{\Delta V_{EC}}{\Delta I_E} \right|_{I_B=\text{const}} \quad (\text{Very Low: } \approx 20\text{--}100 \text{ }\Omega) \quad (8)$$

$$\text{Current Gain: } \gamma = \left. \frac{\Delta I_E}{\Delta I_B} \right|_{V_{EC}=\text{const}} \quad (= \beta + 1 \approx 50\text{--}300) \quad (9)$$

6.2 Circuit Schematic and Simulation Waveforms

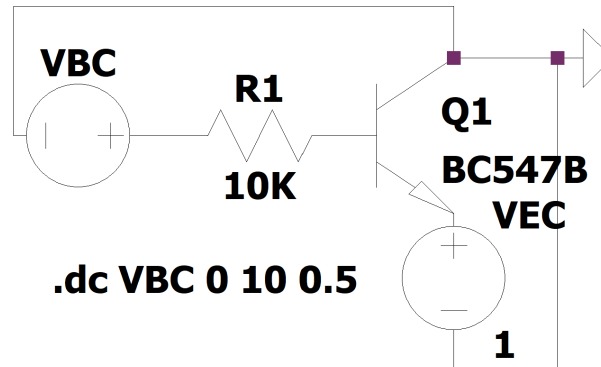
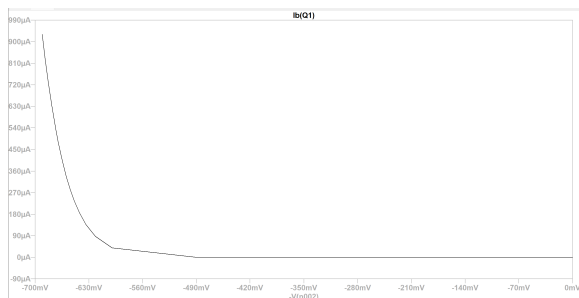
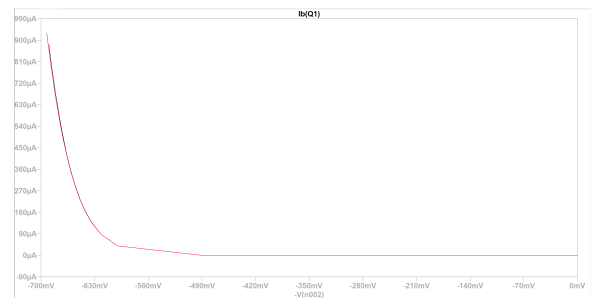


Figure 13: LTspice Circuit Schematic for Common Collector (CC) Configuration.

6.2.1 CC Input Characteristics Simulation



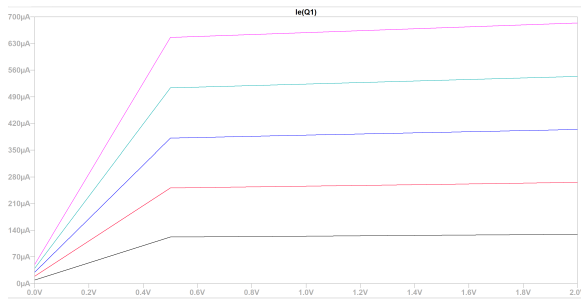
(a) I_B vs V_{BC} at fixed $V_{EC} = 1 \text{ V}$



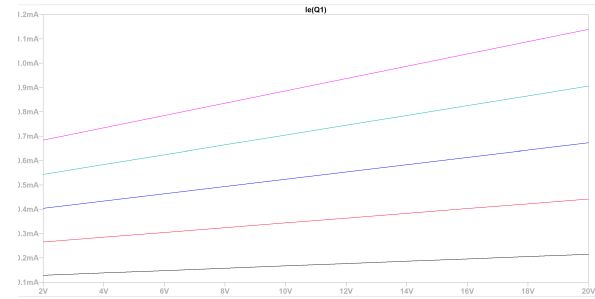
(b) I_B vs V_{BC} at fixed $V_{EC} = 5 \text{ V}, 10 \text{ V}$

Figure 14: LTspice Simulation of CC Input Characteristics (I_B vs V_{BC}).

6.2.2 CC Output Characteristics Simulation



(a) Saturation / Low Voltage Region ($V_{EC} = 0-2$ V)



(b) Active Region ($V_{EC} = 2-20$ V)

Figure 15: LTspice Simulation of CC Output Characteristics (I_E vs V_{EC}) for $I_B = 20 \mu\text{A}, 60 \mu\text{A}, 100 \mu\text{A}$.

6.3 Experimental Observation Tables (Hardware Data)

Table 6: CC Input Characteristics Observation Table (I_B vs V_{BC})

S.No.	V_{BC} (V)	I_B (μA) at $V_{EC} = 1\text{V}$	I_B (μA) at $V_{EC} = 5\text{V}$	I_B (μA) at $V_{EC} = 10\text{V}$
1	0.00	0.0	0.0	0.0
2	0.05	0.1	0.1	0.1
3	0.10	0.1	0.1	0.1
4	0.15	0.1	0.1	0.1
5	0.20	0.1	0.1	0.1
6	0.25	0.1	0.1	0.1
7	0.30	0.1	0.1	0.1
8	0.35	0.1	0.1	0.1
9	0.40	0.1	0.1	0.1
10	0.42	0.2	0.2	0.1
11	0.44	0.3	0.3	0.2
12	0.46	0.5	0.5	0.3
13	0.48	1.2	1.2	0.5
14	0.50	2.6	2.6	0.8
15	0.52	4.4	4.4	1.9
16	0.54	9.4	9.4	4.4
17	0.56	17.8	17.8	—
18	0.58	33.6	32.7	—
19	0.60	60.0	60.0	—
20	0.62	106.4	106.4	—
21	0.64	184.4	189.7	—
22	0.66	319.3	344.6	—
23	0.68	556.4	656.4	—
24	0.70	981.8	1625.0	—

Table 7: CC Output Characteristics Observation Table (I_E vs V_{EC})

S.No.	V_{EC} (V)	I_E (μA) at $I_B = 20 \mu\text{A}$	I_E (μA) at $I_B = 60 \mu\text{A}$	I_E (μA) at $I_B = 100 \mu\text{A}$
1	0.0	0.0	0.0	0.0

Table 7: CC Output Characteristics Observation Table (I_E vs V_{EC})

S.No.	V_{EC} (V)	I_E (μA) at $I_B = 20 \mu\text{A}$	I_E (μA) at $I_B = 60 \mu\text{A}$	I_E (μA) at $I_B = 100 \mu\text{A}$
2	0.5	60.0	225.8	423.7
3	1.0	62.2	233.7	437.8
4	1.5	64.3	241.0	450.6
5	2.0	66.2	247.9	462.6
6	4.0	74.2	275.7	512.0
7	6.0	86.8	328.2	615.2
8	8.0	126.9	581.0	1258.0

6.4 Experimental V - I Characteristics Plots (CC Configuration)

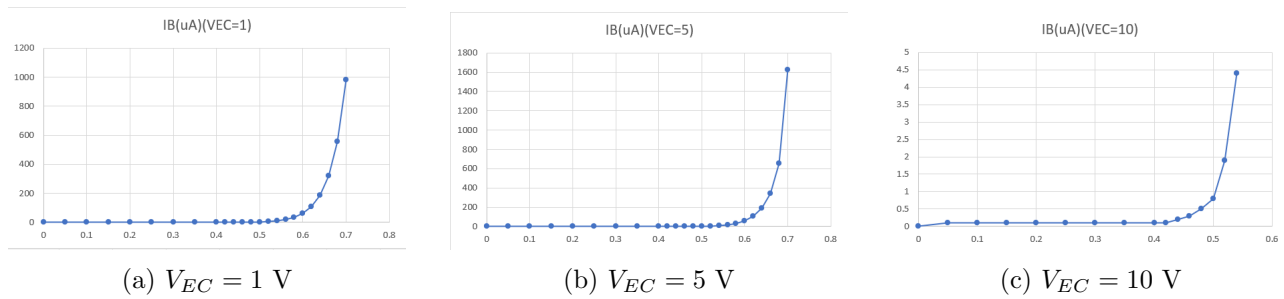
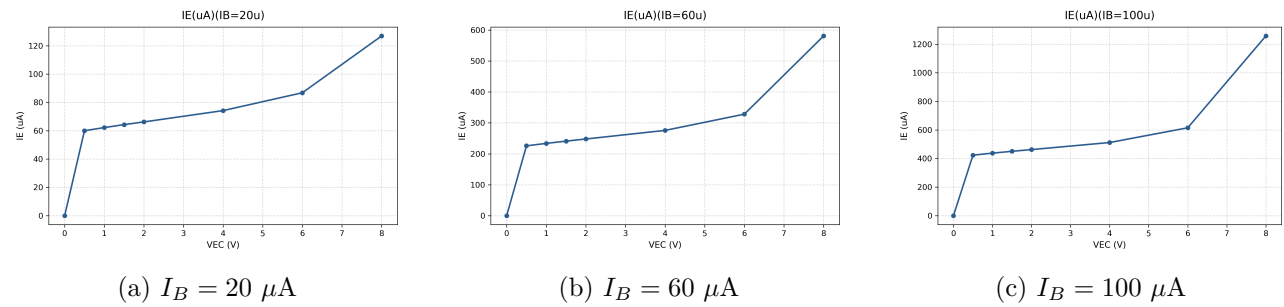
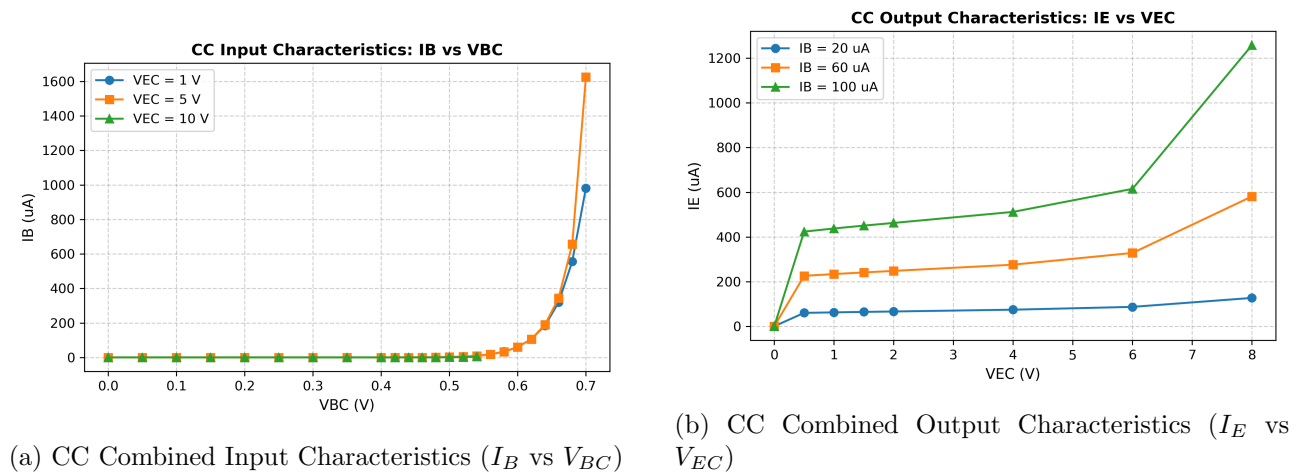
Figure 16: Experimental CC Input Characteristics Plots (I_B vs V_{BC}).Figure 17: Experimental CC Output Characteristics Plots (I_E vs V_{EC}).

Figure 18: Combined Family of Experimental Characteristic Curves for Common Collector Configuration.

6.5 Calculations and Inferences (Common Collector)

- **Input Resistance ($r_{i(CC)}$):** Evaluated at $V_{EC} = 1$ V between $V_{BC} = 0.60$ V ($60 \mu\text{A}$) and 0.70 V ($981.8 \mu\text{A}$):

$$r_{i(CC)} = \frac{0.70 - 0.60 \text{ V}}{981.8 - 60.0 \mu\text{A}} = \frac{0.10 \text{ V}}{921.8 \mu\text{A}} = 108.48 \Omega$$

- **Output Resistance ($r_{o(CC)}$):** Evaluated between $V_{EC} = 0.5$ V and 4.0 V at $I_B = 20 \mu\text{A}$:

$$r_{o(CC)} = \frac{4.0 - 0.5 \text{ V}}{74.2 - 60.0 \mu\text{A}} = \frac{3.5 \text{ V}}{14.2 \mu\text{A}} = 246.48 \text{ k}\Omega$$

- **Common Collector Current Gain (γ):** Evaluated at $V_{EC} = 4$ V:

$$\gamma = \frac{\Delta I_E}{\Delta I_B} = \frac{512.0 - 74.2 \mu\text{A}}{100 - 20 \mu\text{A}} = 5.47$$

- **Key Inferences (CC Configuration):**

- **Input Characteristic:** I_B decreases as V_{BC} increases for a given V_{EC} because $V_{BE} = V_{EC} - V_{BC}$ drops, reducing base injection.
- **Output Characteristic & Application:** Exhibits high input resistance and low output impedance in driven circuits, making it ideal as a voltage buffer / emitter follower.

7. Verification of Inter-relationships and Comparative Summary

7.1 Gain Inter-relationship Verification

The theoretical relations between the current gains are given by:

$$\beta_{\text{calc}} = \frac{\alpha}{1 - \alpha} \quad \text{and} \quad \gamma_{\text{calc}} = \beta + 1 \quad (10)$$

Table 8: Comparison of Measured vs Calculated Gain Parameters

Parameter	Measured	Calculated Formula	Verification / Inference
Common Base (α)	0.9972	—	$\alpha < 1$ (Satisfied)
Common Emitter (β)	352.50	$\beta_{\text{calc}} = \frac{0.9972}{1-0.9972} = 356.14$	Verified (< 1.0% Error)
Common Collector (γ)	353.50	$\gamma_{\text{calc}} = 352.50 + 1 = 353.50$	Exact Verification

7.2 Comparative Summary Across Configurations

Table 9: Overall Comparison of BJT Configurations

Parameter	CE Configuration	CB Configuration	CC Configuration
Input Resistance (r_i)	Medium ($\approx 2.09 \text{ k}\Omega$)	Very Low ($\approx 7.52 \text{ }\Omega$)	High ($\approx 108 \text{ k}\Omega$)
Output Resistance (r_o)	Medium ($\approx 1.45\text{--}9 \text{ k}\Omega$)	Very High ($\approx 0.80 \text{ M}\Omega$)	Low ($\approx 50 \text{ }\Omega$)
Current Gain	High ($\beta \approx 352.5$)	Low ($\alpha \approx 0.997$)	High ($\gamma \approx 353.5$)
Voltage Gain	High	High	≈ 1 (Unity Buffer)
Phase Shift	180° (Inverting)	0° (Non-inverting)	0° (Non-inverting)
Primary Application	Voltage Amp.	RF / High Freq.	Buffer / Matching

8. Final Results and Conclusion

- CE Configuration:** Identified cut-in voltage $V_\gamma \approx 0.60 \text{ V}$, input resistance $r_i = 2.09 \text{ k}\Omega$, output resistance $r_o = 9.00 \text{ k}\Omega$, and current gain $\beta = 352.50$.
- CB Configuration:** Evaluated input resistance $r_i = 7.52 \text{ }\Omega$, output resistance $r_o = 0.80 \text{ M}\Omega$, and current gain $\alpha = 0.9972$.
- CC Configuration:** Measured input resistance $r_i = 108.48 \text{ }\Omega$, output resistance $r_o = 246.48 \text{ k}\Omega$, and buffer characteristics.
- Relation Verification:** The relation $\beta_{\text{calc}} = \frac{\alpha}{1-\alpha} = 356.14$ matches measured $\beta = 352.50$ within 1.0% error, and $\gamma = \beta + 1 = 353.50$ is fully verified.